Human Learning as Hierarchical Pattern Modelling

Matt Thomson. imatt.au@protonmail.com

Working Paper v1.25 (with clarifications, revised Section 2, and minor precision updates)
Working paper note

This is a working paper. It sets out an initial formulation of the hierarchical pattern modelling framework and is intended to invite critique, refinement and empirical testing. The testable predictions in Section 9 are provisional. They indicate how the framework could, in principle, be constrained or falsified rather than providing fully specified experimental designs.

AI Assistance Disclosure

This working paper was developed with substantial assistance from large language models (LLMs), including ChatGPT and Gemini. These systems were used for drafting text, generating alternative explanations, checking logic consistency, proposing mathematical formulations, and proofreading. All conceptual framing, hypothesis development, empirical predictions, structural decisions and revisions were directed, reviewed and approved by the human author. The author takes full responsibility for the content, claims and any errors.

1. Introduction

Human learning is often described as a mystery. Research provides detailed accounts of specific processes such as reinforcement learning, memory consolidation, concept formation and language acquisition. However, there is still no simple and unifying explanation that links these processes into a coherent picture of cognition. This paper proposes a unifying hypothesis.

Human learning is the progressive construction and refinement of hierarchical patterns that capture structure in the world, the body and social life. Any learning episode can be described as a change in how a system represents, predicts and evaluates structure across multiple levels of abstraction. Patterns are treated as deep structural invariances rather than surface correlations. This allows perception, memory, reasoning, imagination, creativity and social learning to be interpreted as variations of a common activity. discovering, stabilising, manipulating and recombining structure.

The account does not deny the importance of specific mechanisms, such as synaptic plasticity, Bayesian inference or reinforcement learning. Instead, it interprets these mechanisms as different ways biological systems discover, maintain and use hierarchical patterns under constraints of energy, time, risk and social context.

Section 2 outlines the core hypothesis. Section 3 illustrates its scope, Section 4 discusses

evolution and truth, Section 5 explains superstition and science, Section 6 identifies challenges, Section 7 highlights implications for learning, Section 8 examines implications for artificial intelligence and Section 9 offers testable predictions.

This paper presents hierarchical pattern modelling (HPM) as a unifying hypothesis about human learning and cognition. It does not aim to replace existing mechanistic theories, nor to prescribe a single computational architecture.

Instead, it offers a conceptual framework that brings together diverse findings across perception, memory, reasoning, creativity, development and social learning under a shared organising principle. A unifying hypothesis operates at a different level to a mechanistic model.

Its purpose is to identify common structural elements across seemingly distinct cognitive processes and to propose a coherent vocabulary that can integrate multiple explanations, rather than to specify the neural or algorithmic details of those processes.

In this sense, HPM plays the same role that early formulations of natural selection, the germ hypothesis or the predictive processing framework played in their initial stages, establishing a generative conceptual scaffold that subsequent theories can formalise and test.

HPM therefore makes the following deliberately bounded claims:

1. Learning involves the discovery, stabilisation and refinement of patterns across multiple substrates, scales and levels of abstraction.
2. The framework is compatible with existing cognitive mechanisms such as reinforcement learning, predictive processing, Bayesian inference, schema formation and associative learning.
3. HPM is not a new cognitive architecture. It is a higher-level conceptual lens through which these mechanisms can be interpreted and connected.
4. The value of the hypothesis lies in unification: it reveals why very different cognitive phenomena, from sensory learning to creativity, from superstition to scientific reasoning, can be understood as variations of a common pattern-modelling process.
5. The paper does not claim that every detail is fully specified. Sections 6 and Appendix A explicitly outline open questions, constraints and directions for future formalisation.
6. HPM is empirically constrainable. Section 9 provides testable predictions and clear counter-cases that would challenge or falsify key claims, ensuring the hypothesis does not collapse into an unfalsifiable generality.

By framing HPM explicitly as a unifying hypothesis, rather than as a mechanistic proposal or a competitor to existing cognitive theories, this paper situates its contribution in conceptual clarity, integrative power and testable structure. Mechanistic precision is left to future work, and the hypothesis is intentionally broad in order to map the deep commonalities across diverse domains of human cognition.

2. The pattern hypothesis

2.1 Core claim

The central claim is that human learning is best understood as hierarchical pattern modelling. A pattern is a stable structure that can be used to compress experience, predict what comes next, guide action or support simulation of alternatives. Learning involves discovering such structure, encoding it in internal and external substrates and continually refining it in light of new evidence and evaluative pressures. Perception, memory, concept formation, language, reasoning, planning and imagination can all be understood as operations over patterns.

2.2 Broad definition of pattern

A pattern is a structured regularity. It is a stable relation across elements of experience, action or representation.

Patterns include:

- spatial and temporal invariances
- latent structural features such as objecthood, agency and constraint
- causal and conditional relations
- relational structures such as dependency, hierarchy and transformation
- abstract regularities such as number, fairness and intention
- internal strategies, errors and metacognitive structures
- counterfactual models describing how systems could behave

Patterns are not restricted to sensory surfaces. They can exist in bodily states, in motor routines, in symbolic media, in social norms and rituals and in institutions. Clarifying the status of patterns

A pattern does not need to be objectively accurate to function as a pattern for a learner. What matters is that the learner treats some structure as stable enough to guide prediction, action, explanation or simulation. A pattern may therefore be veridical, partially correct or entirely spurious. The cognitive system does not inherently discriminate between these cases. It stabilises whichever structures persist under the influence of sensory data, evaluators and the surrounding pattern field. To avoid circularity, a representation qualifies as a pattern only if it satisfies three criteria.

1. It has internal coherence.
2. It supports prediction, explanation, action or mental simulation.
3. It is reinforced or maintained by evaluators such as affect, curiosity, reward, identity or social feedback.

Under this definition, illusions, superstitions, stereotypes and misinformation can all become patterns when they exhibit coherence, contextual utility and evaluator support.

The distinction between accurate and inaccurate patterns arises from the quality of environmental input, the structure of evaluators and the design of the social and institutional pattern field. The same architecture can therefore explain both accurate structural learning and stable but incorrect beliefs.

2.3 Hierarchical pattern formation

Learning involves multiple levels of structure. Consider a child learning to catch a ball. Over time they acquire:

1. Sensory regularities. Regular patterns in how objects move across the visual field or feel in the hand.
2. Latent structural representations. Expectations about persistence, gravity and motion.
3. Relational rules. Relations between speed, angle and distance.
4. Generative rules. The ability to mentally simulate new throws or positions.
5. Meta patterns. Strategies for monitoring success, correcting errors and evaluating performance.

These levels form a hierarchy. Lower level patterns feed into more abstract structure, which supports generative modelling, which is monitored by meta level patterns. Hierarchical pattern formation is not a rigid ladder. Rather, it is a loose description of how structure tends to accumulate and become more abstract over time, given repeated interaction with the world.

2.4 Innate structure and evolutionary priors

The hypothesis does not treat human learners as blank slates. Biological evolution has already discovered powerful patterns about the structure of the world and built them into nervous systems and bodies. These include:
biases in perception, such as edge detection, motion sensitivity and face preference
predispositions for learning certain patterns, such as phonetic contrasts, social contingencies and basic numerosity
affective responses that privilege certain outcomes, such as pain avoidance, novelty seeking and affiliation

Evolution optimises for survival and reproduction, not for truth. It provides a starting architecture and a set of priors that make some patterns easier to learn than others. Individual learning then operates over these priors. It refines them when conditions allow, and in many cases it builds further abstractions on top of them. There is no guarantee that these learned patterns approximate objective truth. They are evaluated primarily by behavioural adequacy, not by correspondence to an independent reality.

At the same time, humans have created cultural and institutional mechanisms that can, under some conditions, increase the alignment between learned patterns and external structure. The scientific method is one prominent example, discussed further in Section 5.

2.5 Pattern substrates, dynamics, evaluators and fields

“Pattern” can easily become an empty label if everything counts as a pattern. To avoid this, the framework distinguishes four roles in any learning system:

1. Pattern substrates
2. Pattern dynamics
3. Pattern evaluators and gatekeepers
4. Pattern fields and institutions

These roles allow us to talk about where patterns live, how they change, what selects them and what wider environments they inhabit.

2.5.1 Pattern substrates

Pattern substrates are the media in which patterns are encoded. For humans, these include:
neural activity and synaptic structure
bodily states, posture and movement
external symbolic systems such as spoken and written language, diagrams, maps, mathematical notation and code
artefacts and tools, from notebooks and smartphones to software systems
social practices, rituals, narratives, roles and norms

Learning often involves shifting patterns between substrates. A skill may begin as an explicit verbal rule, become a visual diagram, then settle into a largely tacit motor routine. A law may begin as a social practice, become a written statute and finally be encoded into software.

2.5.2 Pattern dynamics

Pattern dynamics describe how patterns are created, updated, stabilised and lost. They include:
associative mechanisms that strengthen co occurrence
prediction error driven learning, where discrepancies between expectation and outcome drive change
reinforcement and punishment, where outcomes shift the probability of reusing a pattern
structural integration, where patterns are combined into larger units or decomposed into subpatterns
forgetting, interference and decay

Different domains emphasise different dynamics. Language learning relies heavily on distributional statistics and social feedback. Motor learning relies heavily on practice, error correction and bodily feedback. Social learning relies on observing others, copying and selective imitation.

2.5.3 Pattern evaluators and gatekeepers

Not all learned patterns stabilise. Some are reinforced and integrated, while others remain fleeting or are actively suppressed. Pattern evaluators are the systems that decide which patterns are worth keeping. These include:
affective systems, such as fear, pleasure and shame
curiosity and boredom mechanisms that drive exploration or withdrawal
social evaluators, such as approval, status, punishment and reward
internal standards, such as values, identity and coherence requirements

Evaluators act as gatekeepers. They bias attention to certain patterns, influence which ones are rehearsed, and shape how conflicts between patterns are resolved.

For example, a child may discover a pattern that teasing classmates produces laughter and

attention, which is rewarding. The same child may also learn that this behaviour conflicts with internalised norms of kindness. The resulting tension between evaluators can drive further learning, such as seeking alternative ways to gain attention.

2.5.4 Pattern fields and institutions

Patterns do not live only in individuals. They inhabit broader pattern fields. networks of practices, artefacts, norms and institutions that make some patterns highly available and others rare or costly to maintain. Examples include:
families and peer groups
media ecosystems and online platforms
workplaces, professional cultures and bureaucracies
legal systems, religions, educational systems and scientific communities

These pattern fields provide structured streams of data and feedback that shape what individuals can easily learn. For instance, a media ecosystem that amplifies emotionally charged but misleading content creates a pattern field in which certain spurious correlations are far easier to acquire and maintain than others.

Science, on this view, is a deliberately engineered pattern field. It attempts to increase data quality, support replication, enforce shared notations and punish certain classes of error. This makes some structural patterns easier to stabilise and others harder to maintain.

2.6 Pattern density and pattern stickiness

Some patterns become unusually stable. They persist across contexts, resist contradictory feedback and attract repeated use. Hierarchical Pattern Modelling interprets this stability as the result of pattern density. Pattern density refers to the degree to which a pattern is internally interconnected, evaluator rich and supported by the surrounding pattern field.

A pattern has high density when it satisfies three conditions.

1. Internal structural connectivity
A dense pattern contains many mutually reinforcing subpatterns. These subpatterns provide redundant support so that local errors have little effect on the overall structure. High connectivity reduces susceptibility to rapid decay. This explains why a false pattern can remain stable even when people encounter disconfirming evidence. Stability arises from internal linkage rather than epistemic quality alone.

2. Evaluator saturation
Multiple evaluators reinforce the same pattern. Identity, emotion, curiosity, coherence, social reward and task utility can all act to preserve a structure. When evaluators converge, the pattern receives repeated reinforcement and is reused frequently. Evaluator saturation helps explain why some inaccurate or simplified patterns remain attractive: they satisfy several evaluators simultaneously.

3. Field amplification
Many patterns are stabilised by the larger pattern fields they inhabit. These include media

ecosystems, educational environments, cultural narratives and institutional routines. A pattern that is repeatedly encountered through these fields gains activation energy from the environment. This allows patterns to persist even when they lack strong predictive accuracy or direct personal experience.

Pattern density explains why structure can be stable without being accurate. Density describes stability, not truth. A pattern may be dense because it compresses many experiences into a manageable representation, because it is socially rewarded or because its components are frequently reused.

This framework predicts that interventions that reduce density, such as weakening evaluators or altering pattern fields, should make sticky patterns easier to revise. Conversely, teaching methods that increase density around deep structure should improve long-term learning and transfer. Pattern density therefore links micro-level cognitive processes to broader environmental and institutional influences on learning.

3. From perception to abstract thought

Human learning scales from low level sensory patterns to high level symbolic, conceptual and imaginative patterns. The same underlying process operates across these levels. the discovery, refinement and recombination of structure.

3.1 Perceptual pattern formation

At the perceptual level, sensory systems learn to track regularities in light, sound, touch and internal bodily signals. They detect invariances under rotation, translation and noise. They segment objects, track motion and identify boundaries. These patterns are tightly coupled to sensorimotor experience and provide the foundation for higher levels of structure.

3.2 Intermediate relational structure

At intermediate levels, patterns capture relations among objects and events. These include regularities such as who acts upon whom, under what conditions, with what consequences. Such patterns support basic causal reasoning, prediction and simple planning. They allow the learner to build expectations that generalise across contexts and to anticipate how systems behave under different circumstances.

3.3 Higher level conceptual and symbolic patterns

At higher levels, patterns become conceptual, symbolic and narrative. People learn patterns of argument, story structure, social roles, institutional rules and moral principles. These patterns describe not only what happens, but what could or should happen. They support counterfactual reasoning and long range planning.

Abstract thought emerges when the learner can manipulate these higher level patterns in loosely decoupled ways. Internal models are used to run simulated scenarios, evaluate options and update beliefs. This process allows new structures to be explored without immediate dependence on external feedback.

3.4 Creativity as pattern recombination

Imagination and creativity arise from the recombination of stored patterns under learned constraints. A creative idea is a structural recombination that preserves deeper invariances while breaking surface expectations. Creative thinking does not require a separate cognitive faculty. It is a mode of pattern exploration in which existing structures are rearranged into novel but coherent forms. The distinction between routine and creative thought lies in the degree of structural transformation. Routine cognition applies well learned patterns to familiar situations. Creative cognition recombines patterns across levels, domains or evaluative contexts.

3.4.1 The phenomenology of originality and insight

Creative insight often feels sudden or as if it arises from nowhere. Under Hierarchical Pattern Modelling, this phenomenology arises when previously unconnected patterns become jointly active and recombine into a new, more coherent structure. When the newly formed pattern sharply reduces prediction error or increases evaluator scores, the resulting change in the total score is registered as a salient shift.

Some learners experience frequent high salience insights. Others experience creativity as gradual refinement, as patterns are incrementally adjusted until prediction, evaluator signals and structural coherence improve enough for the change to be noticed. In both cases, what feels like “something out of nothing” is the system reconfiguring existing patterns into a configuration that was previously inaccessible.

Individual differences in creativity follow naturally. Variations in evaluator settings, substrates and pattern fields change how often pattern recombinations occur and how strongly they are reinforced. This covers differences in tolerance for uncertainty, reward for novelty, associative richness, working memory and the cultural or material tools that allow patterns to be externalised and recombined.

3.5 Metacognitive patterns

Metacognition involves recognising patterns in one’s own reasoning, strategies and errors. It includes patterns that evaluate the reliability of other patterns, select among competing approaches and guide self correction. Meta level patterns influence how people monitor progress, allocate attention and adjust their learning strategies over time.

4.Innate structure, evolution and truth

As noted in Section 2.4, biological evolution provides the initial architecture and priors for

learning. It gives organisms a starting set of pattern detectors and action policies that have historically yielded fitness.

This raises an important question. if human learning is fundamentally pattern based and evolution optimises for survival rather than truth, how do humans ever arrive at reasonably accurate models of the world?

The framework suggests a layered answer.

1. Behavioural adequacy as the primary criterion In everyday life, learned patterns are selected primarily for their usefulness in guiding action under local constraints. Many adequate patterns are partly wrong or parochial, but still good enough for local purposes.
2. Informational environments and pattern fields The quality of patterns that individuals can learn depends strongly on the structure of their informational environment. Noise, bias, censorship and algorithmic amplification can all distort which patterns are salient and which feedback signals are available.
3. Meta level pattern refinement Cultural practices such as mathematics, controlled experiment, peer review and rule of law aim to improve feedback quality, reduce some biases and support more reliable pattern extraction. They are meta patterns that shape how other patterns are evaluated and stored. In short, humans do not naturally learn “truth”. They learn patterns that have been filtered by evolutionary, affective, social and institutional evaluators. Under some conditions, especially in well designed pattern fields, these patterns track deep structure reasonably well. Under others, they can stabilise serious systematic errors.

5. Superstition, misinformation and science

The framework provides a way to describe superstition, misinformation and science within a single vocabulary.

5.1 Spurious patterns

Humans are excellent pattern finders. The same talent that allows us to detect real structure in noisy data also makes us vulnerable to spurious correlations. Superstition can be seen as the stabilisation of spurious patterns under conditions of high affect, low feedback quality or high noise.

For example, a gambler who wins after wearing a particular shirt may form a pattern linking shirt and success, especially if the event is emotionally charged and if feedback about randomness is poor. Once a spurious pattern has been encoded, evaluative systems may preserve it. Confirmation bias leads people to notice supporting cases more than disconfirming ones. Social reinforcement may reward storytelling around the pattern. Over time, the pattern can become entrenched, even in the face of contradictory evidence.

5.2 Distorted pattern fields

Modern information ecosystems can be understood as pattern fields that selectively amplify certain structures. Recommendation systems, advertising models and social sharing dynamics often favour content that is novel, emotionally arousing or identity affirming. Such systems may, without any malicious intent, create environments where certain misleading patterns are far more visible, repeated and socially rewarded than more accurate but less engaging ones. Misinformation, in this sense, is not only about individual gullibility. It is about the design of pattern fields that make some patterns very easy to learn and others very hard.

5.3 Science as a pattern refinement institution

Science can be interpreted as a deliberately engineered set of pattern dynamics and evaluators. It attempts to:
improve data quality, for example through careful measurement, sampling and control
support replication and cumulative refinement
enforce shared notations and standards for argument
reward certain kinds of correction and penalise certain kinds of error

Under this view, science does not guarantee truth. Rather, it constructs a pattern field in which patterns that track real structure tend to be more stable than those that do not, at least over time and across many contributors. Superstition, misinformation and science thus become different regimes of pattern learning. They differ in informational quality, evaluative pressures and institutional design, not in the basic fact that they involve stabilising patterns.

6. Challenges and open questions

The hierarchical pattern modelling hypothesis faces several important challenges. Only a few are noted here.

1. Formal precision The framework is presented at a conceptual level. It needs more formal models that specify how patterns are represented, updated and selected in computational terms.
2. Granularity and levels The notion of “hierarchy” in pattern formation is intentionally broad. It must be clarified in relation to existing multi level models in neuroscience, psychology and AI, and linked to measurable quantities.
3. Resource constraints Real learners face severe limits on time, energy, data and computation. How do they approximate hierarchical pattern learning under these constraints, and how does this shape what they can and cannot learn?
4. Developmental trajectories Human learning unfolds over long developmental timescales, from infancy into adulthood. The framework must be tested against known developmental milestones and constraints.
5. Individual and cultural variation Different individuals and cultures stabilise different patterns, even under similar environments. The framework must account for this variation, including differences in evaluators and pattern fields.
6. Discriminating from neighbouring theories Many current approaches, such as predictive processing, Bayesian models and connectionist frameworks, also emphasise learning and

structure. The pattern hypothesis needs testable predictions that distinguish it from these neighbours. Section 9 addresses the final point by outlining specific predictions and contrasts.

7. Implications for understanding human learning

The framework offers a unified way to think about several aspects of human learning. Only three are sketched here.

7.1 Expertise

Expertise can be seen as the accumulation of dense, compressed pattern structure in a domain. Experts do not simply remember more isolated facts. They perceive higher order regularities, recognise familiar structures in novel surface forms and can recombine patterns flexibly. On this view, developing expertise involves:
long exposure to rich, structured examples
feedback that highlights relevant invariances and suppresses misleading correlations
opportunities to externalise, reflect on and revise pattern structures, for example through diagrams, explanations and teaching

7.2 Bias and illusion

Cognitive biases and illusions can be interpreted as stable but misaligned patterns. They are patterns that were once adaptive, that are adaptive in some environments but not others, or that are preserved by strong evaluators such as affect and social identity. For example:
confirmation bias can be seen as a pattern in how evidence is sampled and weighted
illusions in perception can be seen as side effects of priors that are usually useful but misfire in specific contexts
stereotypes can be seen as overgeneralised patterns learned in skewed social environments
Addressing bias is not simply a matter of “correcting beliefs”. It involves altering pattern fields, evaluators and dynamics so that alternative patterns can stabilise.

7.3 Education

If learning is hierarchical pattern modelling, education is the deliberate design of pattern fields and evaluative environments. Effective education should:
expose learners to rich examples that share deep structure but vary at the surface
make key patterns explicit through comparison, contrast and representation
provide feedback that rewards discovery of real structure and discourages overfitting to superficial features
support externalisation of patterns in language, diagrams, models and collaborative work
This perspective emphasises far transfer. The aim is not only performance on tasks that

resemble training examples, but competence in applying learned patterns to structurally related but superficially different problems.

7.4 Developmental Pattern Progression

Hierarchical Pattern Modelling implies that learners do not begin with access to deep or abstract structure. Instead, they stabilise progressively more complex patterns over time. This progression reflects both cognitive constraints and the cumulative nature of pattern formation. The following levels outline a tentative developmental trajectory. They represent increasingly abstract forms of structure, not sharp stages. Individual learners may progress unevenly depending on prior knowledge, evaluator settings, instructional support and domain familiarity.

7.4.1 Level 1. Surface-Feature Patterns (Novice Stage)

Early learning involves stabilising simple, concrete regularities with minimal abstraction. These include perceptual cues, basic associations and procedural steps.
Examples include:
letter–sound correspondences
counting routines
simple action–outcome relations
recognition of common visual or linguistic patterns

These patterns form quickly but are fragile and highly context-dependent. Cognitive load theory predicts that novices rely heavily on these small, low-complexity units because they require minimal working memory.

7.4.2 Level 2. Local Structural Patterns

As surface patterns accumulate, learners begin to detect slightly deeper structure within a domain. These patterns involve relationships among surface features but remain concrete and context-bound. Examples include:
“bigger denominators often imply smaller fractions”
“stories usually involve a problem and a resolution”
“objects accelerate when pushed harder”

These patterns show the beginnings of generalisation but still depend on familiar surface forms.

7.4.3 Level 3. Relational Patterns

With further exposure and support, learners begin to encode relationships that generalise across multiple contexts. This level introduces genuine structure detection.
Examples include:
proportionality
conservation relations
conditional dependencies
simple causal schemas
At this level, learners can apply patterns beyond the exact examples used in instruction. Far transfer becomes possible but is still fragile.

7.4.4 Level 4. Abstract Structural Patterns

Advanced learners compress multiple relational patterns into higher-level schemas. These patterns are less tied to perceptual detail and allow for flexible reasoning. Examples include:
linear functions and rate-of-change concepts
grammar as a system of constraints
narrative templates
Newtonian principles such as inertia or superposition

These patterns can be recombined and manipulated internally without direct dependence on surface cues.

7.4.5 Level 5. Generative Patterns (Expert Stage)

Expertise involves the ability to generate new patterns by recombining or transforming existing structure. Experts can:
design new models
construct novel explanations
recognise deep similarity across superficially unrelated problems
adapt patterns rapidly to new situations

Expert performance reflects dense, compressed, multi-level pattern organisation.

7.4.6 Instructional Implications

This progression suggests that teaching should be matched to the learner's current pattern complexity.

1. Novices require explicit instruction, simplified examples, worked solutions and clear surface cues. They cannot reliably extract deep structure without scaffolding.
2. Intermediate learners benefit from contrasting cases, structured variation and tools that externalise relational information.
3. Advanced learners require opportunities for abstraction, explanation, modelling and deliberate practice across varied contexts.
4. Experts-in-training need generative tasks that encourage recombination and transformation of known patterns. Instruction that introduces structural complexity too early risks stabilising incorrect patterns or overwhelming working memory. Conversely, instruction that remains at surface levels may prevent learners from reaching relational or abstract structure.

7.4.7 Boundary Conditions

This progression should not be interpreted as a strict sequence. Limitations include:

- domain differences in the availability of hierarchical structure
- individual variation in cognitive capacity, motivation and evaluators

- social factors that accelerate or inhibit pattern progression
- cases where the instructional priority is the stabilisation of procedural patterns rather than explicit abstract understanding

More empirical work is needed to determine how these levels emerge, how they vary across domains and whether they can be measured reliably.

8. Implications for artificial intelligence

The framework is not a direct proposal for AI architectures, but it does suggest some directions.

8.1 Beyond correlation

Large models trained on massive datasets can capture many patterns in data. However, they may primarily encode surface correlations unless their training regimes and objectives encourage the discovery of deeper structure. A pattern based perspective draws attention to the need for:
objectives that reward structural fidelity and counterfactual robustness
training curricula that emphasise variation at the surface while preserving underlying structure
mechanisms for representing and manipulating explicit structural patterns, not only distributed correlations

8.2 Hierarchical generative modelling

Many current AI systems already use hierarchical generative models. The framework supports this direction but stresses the role of evaluators and fields. It suggests that high performing systems may need:
multiple levels of pattern representation, from low level sensory features to high level abstract rules
mechanisms for maintaining and revising generative models under noisy and partial feedback
external pattern substrates, such as tools and notations, that can be used to extend internal models

8.3 Causality and intervention

If learning is about structural pattern discovery, causality becomes central. Systems that only encode observational correlations may fail when interventions change the data generating process. The framework supports ongoing work that incorporates causal structure into learning and planning. It also highlights that causal learning is not only a matter of algorithms. It depends on evaluators and pattern fields that reward interventions and experiments. 8.4 Compositionality and creativity

Artificial creativity can be framed as recombination of structural patterns under constraints. Systems that explicitly represent and manipulate compositional structures may be better able to generate novel yet coherent outputs than systems that rely solely on surface level statistics. 8.5 Human compatible AI

Alignment can be seen as a problem of pattern sharing and pattern evaluation. Human compatible AI would require:

mechanisms for sharing, inspecting and negotiating pattern structures and evaluators between humans and machines

pattern fields that encourage convergence on shared structural models of goals, constraints and acceptable behaviour

institutional designs that can correct and refine misaligned patterns over time

9. Testable predictions and research directions

The hypothesis of human learning as hierarchical pattern modelling is only useful if it can be constrained and, in principle, falsified. This section outlines several testable predictions across different domains. Each prediction links a core claim of the framework to observable outcomes and contrasts these with what a simpler or competing view might expect. These predictions are intentionally high level. They are intended to guide more detailed experimental designs, simulations and field studies rather than to prescribe specific methods.

9.1 Hierarchical structure versus surface features

The framework claims that learning builds hierarchical patterns that are more sensitive to changes in deep structure than to changes in surface features, once overall difficulty is equated. Consider a task with an underlying relational structure, such as an artificial grammar, a simple game or a category system. Changes to the underlying structure that keep surface features similar should have larger effects on performance than changes to surface details that preserve the latent structure.

Systematic evidence that learners are equally sensitive to deep and shallow perturbations would count against a strong hierarchical claim. This should hold even when overall difficulty is matched.

9.2 Expertise as compression and cross surface transfer

The framework treats expertise as dense, compressed patterning across a domain, rather than just more frequent associations. If this is correct, then compared with novices, experts should:

recognise and predict structured configurations more quickly, even when surface details are altered

show evidence of more compressed internal representations, for example needing fewer latent dimensions to account for their similarity judgements or recall patterns

transfer more effectively when the same deep structure is presented in a different surface form

If expertise were mainly a matter of accumulating more flat associations on familiar stimuli, improvements would be largest on trained configurations and weaker for cross surface transfer. A lack of compression or transfer effects, once practice amount is controlled, would challenge the stronger claims about hierarchical pattern construction.

9.3 Affective evaluators and spurious pattern stabilisation

The framework assigns evaluative roles to affective systems. Under uncertainty, emotional salience is expected to bias which patterns stabilise. If this is correct, then in probabilistic or noisy learning tasks:
conditions that heighten affective load, for example threat framing, social status cues or personally meaningful stakes, should produce more persistent endorsement of spurious patterns and greater resistance to disconfirming feedback
neutral conditions with matched information should allow faster pruning of incorrect patterns once feedback contradicts them

A purely accuracy driven prediction error account might expect affect mainly to alter overall learning rate or exploration, not selectively to stabilise incorrect patterns in the presence of clear feedback. Evidence that affective load has no specific effect on the persistence of wrong rules would weaken the claim that affective evaluators play a structural role in pattern stabilisation.

9.4 Curiosity and the interesting middle

The framework treats curiosity as an evaluator that favours regimes where pattern improvement is possible. Neither perfect predictability nor unstructured noise is expected to maximise engagement. If this is correct, then when people can freely choose between environments that differ in structure:
highly predictable environments should quickly become boring and attract less exploration over time
highly random environments with no learnable structure should also be avoided once their randomness is recognised
environments with intermediate, learnable but non obvious structure should attract the most sustained exploration and yield the largest learning gains

Simple novelty seeking accounts often predict strong attraction to high change or high entropy environments, even when they are not learnable. Systematic preference for mid structured environments, along with better performance there, would support the view of curiosity as a pattern improvement evaluator.

9.5 Social pattern fields and convergence

The framework claims that social fields and institutions alter which patterns are available to be learned and how they stabilise. High quality pattern fields should support convergence on shared underlying structure. Distorted fields should amplify charismatic but incorrect patterns. If this is correct, then in controlled micro culture settings where groups explore the same hidden pattern under different information conditions:
groups with structured feedback, simple shared notations and explicit norms for evidence should converge more quickly on the correct pattern and more reliably abandon incorrect ones
groups with noisy or prestige biased feedback, no shared notation and little record keeping

should show slower convergence, more persistent incorrect patterns and larger divergences between groups

Views that focus mainly on individual learning mechanisms would expect differences in social field design to have weaker effects once individual information exposure is equated. Strong and reliable effects of field design on pattern convergence would support the claim that learning operates within pattern fields, not only within individual brains.

9.6 Education as exposure to pattern structure

The framework suggests that effective education engineers exposure to high quality pattern structures, rather than simply increasing the volume of isolated facts. If this is correct, then when different cohorts are taught the same topic under equal time budgets:

- A pattern-first approach that explicitly compares multiple cases, highlights invariances and compresses structure into schemas or diagrams should produce:
  - equal or slightly better performance on near transfer tests that resemble the training examples
  - substantially better performance on far transfer tests that require applying the learned pattern to novel surface forms

- A fact-first approach that focuses on drill and recall without explicit pattern mapping might match or exceed near transfer performance, but should perform worse on measures that require structural generalisation.

A systematic failure to observe far transfer benefits from pattern-centred teaching would challenge one of the central applied claims of the framework.

9.7 Superstition, noise and institutional scaffolding

The framework treats institutions like science as meta-level pattern refinement systems. In noisy environments without such scaffolding, spurious but structured patterns readily stabilize. This generates a testable prediction:

Prediction: In tasks where outcomes are largely random but afford plausible illusory regularities:

1. Individual condition: Participants working alone, without checking tools, will readily form and defend idiosyncratic superstitious rules.

2. Institutional condition: Providing minimal institutional scaffolds (e.g., randomisation checks, peer replication, simple record keeping) will reduce belief in these spurious patterns over time, even when the underlying outcome statistics remain unchanged.

Contrast with competing views: If individual learners could naturally infer randomness after sufficient exposure, such scaffolds would add little benefit. Clear benefits from minimal institutional features would support the claim that reliable pattern refinement is partly an institutional achievement, not just an individual cognitive capacity.

9.8 Summary

This section has outlined how the hierarchical pattern modelling framework can be empirically constrained across multiple levels of analysis:

- Within-person dynamics: Sensitivity to structural vs. surface changes (9.1); affective stabilization of spurious patterns (9.3); curiosity for intermediate complexity (9.4)
- Individual differences: Expertise as pattern compression and cross-surface transfer (9.2)
- Social and institutional levels: Field design effects on pattern convergence (9.5); institutional scaffolding against superstition (9.7)
- Applied contexts: Educational benefits of pattern-centred teaching (9.6)

These predictions demonstrate that the framework is not merely a metaphorical redescription, but generates specific, falsifiable claims. Systematic failure to find these predicted asymmetries and convergence patterns would require revising the stronger claims about hierarchical pattern construction and evaluator-weighted selection.

Future work should refine these high-level predictions into specific experimental designs, computational models, and field studies. The framework's value lies not in being unfalsifiably general, but in providing a generative structure for predicting and explaining diverse learning phenomena. Systematic failure to find the outlined asymmetries and convergence patterns would require revising or abandoning the stronger parts of the hypothesis.

10. Interdisciplinary Relevance

While the framework remains a hypothesis, it has potential implications across several fields. Developmental psychology may find the emphasis on progressive pattern complexity useful for interpreting learning trajectories. Educational researchers may see value in the idea that exposure to structured variation supports transfer. Artificial intelligence and machine learning may consider the distinction between surface correlations and deeper structure as a prompt for alternative training approaches. Clinical and computational psychology may view the pattern–evaluator interaction as a possible way to frame persistent habits or beliefs. Behavioural economics, organisational psychology and sociology may treat the model as a provisional lens for understanding how social norms and information environments influence behaviour.

These connections are tentative, but they indicate where the hypothesis might intersect with existing research programs.

11. Conclusion

This paper proposes that diverse cognitive phenomena can be understood as manifestations of a single underlying activity: hierarchical pattern modelling. On this view, perception, memory, concept formation, reasoning, imagination, expertise, bias and social learning all involve the discovery, stabilisation, manipulation and recombination of patterns across multiple substrates and levels of abstraction.

The framework does not reduce cognition to a single mechanism in the narrow sense. It acknowledges diverse pattern dynamics, evaluators and fields, and it respects the constraints imposed by evolution, development and culture. Its unifying move is to treat all these mechanisms as components of a broader pattern system.

Section 9 sketched initial testable predictions and research directions that can be used to further constrain and, if necessary, falsify the framework. Future work will need to formalise the ideas presented here, connect them more tightly to existing empirical literatures and explore their practical implications for education, institutional design and artificial intelligence.

Appendix A. Minimal mathematical sketch and future directions

This appendix sketches a minimal computational formalisation of the hierarchical pattern modelling hypothesis and outlines future directions for empirical and theoretical work.

The aim is not to fix a single “correct” model. It is to show that the verbal theory can be expressed using standard tools from probabilistic modelling and information theory, in a way that is compatible with multiple concrete implementations.

A.1 Patterns as generative structures

In this framework, a pattern is represented as a generative model hhh that assigns probabilities to streams of experience x1:Tx_{1:T}x1:T:

- x1:Tx_{1:T}x1:T is a sequence of sensory, motor, conceptual or social observations.
- hhh encodes hypotheses about how such sequences are produced.

A simple objective that captures prediction and compression is:

L(h∣x1:T)=−logp(x1:T∣h)+λ C(h)L(h \mid x_{1:T}) = - \log p(x_{1:T} \mid h) + \lambda \, C(h)L(h∣x1:T)=−logp(x1:T∣h)+λC(h)

where:

- −logp(x1:T∣h)- \log p(x_{1:T} \mid h)−logp(x1:T∣h) measures epistemic loss or surprise. Lower is better.
- C(h)C(h)C(h) is a measure of structural complexity or description length.
- λ≥0\lambda \ge 0λ≥0 weights the penalty on unnecessary complexity.

This treats learning as searching over generative models that balance predictive fit against complexity.

A.2 Hierarchical patterns

To capture hierarchical structure, assume each pattern hhh has two layers of latent variables

zt(2)z_t^{(2)}zt(2) and zt(1)z_t^{(1)}zt(1) that generate observations xtx_txt:

zt(2)→zt(1)→xt.z_t^{(2)} \rightarrow z_t^{(1)} \rightarrow x_t.zt(2)→zt(1)→xt.

Let θ(2),θ(12),θ(1)\theta^{(2)}, \theta^{(12)}, \theta^{(1)}θ(2),θ(12),θ(1) parameterise the higher level dynamics, cross level link and observation model respectively. The joint distribution over a trajectory is:

p(x1:T,z1:T(1),z1:T(2)∣θ)=p(x_{1:T}, z_{1:T}^{(1)}, z_{1:T}^{(2)} \mid \theta)
=p(x1:T,z1:T(1),z1:T(2)∣θ)=
p(z1(2)∣θ(2)) p(z1(1)∣z1(2),θ(12)) p(x1∣z1(1),θ(1))\quad p(z_1^{(2)} \mid \theta^{(2)}) \; p(z_1^{(1)} \mid z_1^{(2)}, \theta^{(12)}) \; p(x_1 \mid z_1^{(1)}, \theta^{(1)})p(z1(2)∣θ(2))p(z1(1)∣z1(2),θ(12))p(x1∣z1(1),θ(1))
×∏t=2Tp(zt(2)∣zt−1(2),θ(2)) p(zt(1)∣zt(2),θ(12)) p(xt∣zt(1),θ(1)).\quad \times \prod_{t=2}^T p(z_t^{(2)} \mid z_{t-1}^{(2)}, \theta^{(2)}) \; p(z_t^{(1)} \mid z_t^{(2)}, \theta^{(12)}) \; p(x_t \mid z_t^{(1)}, \theta^{(1)}).×∏t=2Tp(zt(2)∣zt−1(2),θ(2))p(zt(1)∣zt(2),θ(12))p(xt∣zt(1),θ(1)).

This formulation explicitly separates the initial state (t=1)(t = 1)(t=1) from the subsequent transitions (t=2 to T)(t = 2 \text{ to } T)(t=2 to T), avoiding any reference to an undefined initial latent state z0(2)z_0^{(2)}z0(2).

A.3 Loss and compression

For hierarchical models, an approximate posterior qh(z1:T(1),z1:T(2)∣x1:T)q_h(z^{(1)}_{1:T}, z^{(2)}_{1:T} \mid x_{1:T})qh(z1:T(1),z1:T(2)∣x1:T) can be used to define a negative evidence lower bound style loss:

Lhier(h∣x1:T):=Eqh[logqh(z(1),z(2))]−Eqh[logph(x,z(1),z(2))].L_{\text{hier}}(h \mid x_{1:T}) := \mathbb{E}_{q_h}[\log q_h(z^{(1)}, z^{(2)})] - \mathbb{E}_{q_h}[\log p_h(x, z^{(1)}, z^{(2)})].Lhier(h∣x1:T):=Eqh[logqh(z(1),z(2))]−Eqh[logph(x,z(1),z(2))].

Lower values of LhierL_{\text{hier}}Lhier correspond to better hierarchical fit.

A simple compression measure is the mutual information between latent layers under qhq_hqh:

Comp(h):=Hqh[z(1)]−Hqh[z(1)∣z(2)],\text{Comp}(h) := H_{q_h}[z^{(1)}] - H_{q_h}[z^{(1)} \mid z^{(2)}],Comp(h):=Hqh[z(1)]−Hqh[z(1)∣z(2)],

where Hqh[⋅]H_{q_h}[\cdot]Hqh[⋅] denotes entropy under qhq_hqh. High compression indicates that abstract states z(2)z^{(2)}z(2) capture regularities in the more concrete states z(1)z^{(1)}z(1) and observations.

A.4 Evaluators and total objective

In the main text, evaluators capture epistemic, affective and social feedback on a pattern. For a given pattern hhh and history x1:tx_{1:t}x1:t, let:

Epistemic evaluator (accuracy):
Eepi(h∣x1:t):=−logp(x1:t∣h).E_{\text{epi}}(h \mid x_{1:t}) := - \log p(x_{1:t} \mid h).Eepi(h∣x1:t):=−logp(x1:t∣h).
This is already encoded in the loss L(h∣x1:t)L(h \mid x_{1:t})L(h∣x1:t).

Affective evaluator:
Eaff(h∣x1:t),E_{\text{aff}}(h \mid x_{1:t}),Eaff(h∣x1:t),
representing reward, curiosity, comfort or other affective signals.

Social evaluator:
Esoc(h∣x1:t),E_{\text{soc}}(h \mid x_{1:t}),Esoc(h∣x1:t),
representing social approval, norm alignment or other social feedback.

To avoid double counting epistemic quality, the combined evaluator excludes the epistemic term and aggregates only non epistemic evaluators:

J(h∣x1:t):=β Eaff(h∣x1:t)+γ Esoc(h∣x1:t),J(h \mid x_{1:t}) := \beta \, E_{\text{aff}}(h \mid x_{1:t}) + \gamma \, E_{\text{soc}}(h \mid x_{1:t}),J(h∣x1:t):=βEaff(h∣x1:t)+γEsoc(h∣x1:t),

with non negative weights β,γ\beta, \gammaβ,γ. A simple total objective is then:

Total(h∣x1:t):=L(h∣x1:t)−J(h∣x1:t).\text{Total}(h \mid x_{1:t}) := L(h \mid x_{1:t}) - J(h \mid x_{1:t}).Total(h∣x1:t):=L(h∣x1:t)−J(h∣x1:t).

Lower values of Total\text{Total}Total are better. Patterns that achieve low epistemic loss at a given complexity and high evaluator scores are more likely to stabilise. The epistemic contribution appears only through L(h∣x1:t)L(h \mid x_{1:t})L(h∣x1:t), so there is no double counting.

For hierarchical models, the same logic applies with LhierL_{\text{hier}}Lhier and Comp(h)\text{Comp}(h)Comp(h) in place of LLL and an added compression term if desired.

A.5 Relation to the Meta Pattern Rule

Appendix D develops a more explicit dynamical system for pattern weights wi(t)w_i(t)wi(t) based on a total score Totali(t)\text{Total}_i(t)Totali(t). The sketch in this appendix can be linked to that system in at least two ways.

- A minimisation view, where patterns with lower Total(h)\text{Total}(h)Total(h) are more likely to be selected, and the Meta Pattern Rule is treated as a stochastic descent process on this objective.

- A maximisation view, where one uses a score of the form −L+Comp+J-L + \text{Comp} + J−L+Comp+J as in Appendix D, and the replicator dynamics amplify patterns with higher scores.

Both views are compatible with the hypothesis that learning involves evaluator weighted selection over generative patterns. The choice depends on modelling convenience and the desired mapping to empirical data.

A.6 Scope and limitations

The sketch above is deliberately minimal. It omits:

- explicit structural change in the space of patterns

- detailed modelling of evaluator dynamics
- forgetting and interference
- institutional pattern fields and multi agent structure
- resource constraints and developmental staging

The goal is not to claim that human learning literally optimises these exact objectives. Rather, it is to show that the verbal claims about patterns, evaluators and fields can be framed in terms of familiar optimisation and inference tools.

A.7 Future directions

Several directions follow from this minimal sketch.

A.7.1 Structure of patterns

Further work should clarify the representational format of patterns, including:

- probabilistic graphs and factor graphs
- program like generative rules
- relational or causal schemas
- neural or vector space representations with explicit structure.

Different substrates may be more or less suitable for capturing counterfactuals, long range dependencies and multi agent structure.

A.7.2 Evaluator dynamics

Evaluators require deeper modelling. Key questions include:

- how affective and epistemic signals interact in practice
- how evaluators shape attention, exploration and pruning
- how social evaluators generate shared or divergent patterns across groups
- how evaluator settings change over development, stress or institutional context.

Understanding evaluator dynamics could link learning, motivation, emotion and social cognition within a single pattern based account.

A.7.3 Experimental tests

The formalism supports more precise versions of the testable predictions outlined in the main text, for example:

- distinguishing tasks that mainly tune existing structure from tasks that genuinely alter hierarchy
- measuring “pattern field” properties (diversity, redundancy, conflict) and relating them to learning trajectories
- probing how evaluator manipulations (reward structure, social feedback, framing) alter which patterns stabilise.

Designing such experiments would require collaboration with developmental, cognitive and social psychologists.

A.7.4 Pattern fields and institutions

At the institutional level, the same language can be used to describe:

- how media environments, curricula and organisational routines define pattern fields
- how these fields stabilise or destabilise particular evaluators and pattern structures in populations
- how interventions such as changed feedback channels, incentives or exemplars shift the pattern landscape.

Formal models here would likely combine generative learning at the individual level with network and game theoretic models at the group level.

A.7.5 Applications to artificial intelligence

For artificial intelligence, the sketch suggests:

- framing training data and objective design as the engineering of pattern fields and evaluators
- exploring architectures that support explicit structural revision, not just parameter updates
- integrating richer evaluator signals, including curiosity, risk and social alignment objectives.

The goal is not to force AI systems into a human mould, but to use the hierarchical pattern perspective to highlight blind spots in current practice and to suggest alternative training regimes.

A.7.6 Developmental and resource constraints

Finally, the framework must incorporate constraints that characterise real learners, including:

- limited data and computation
- developmental staging of representational capacity
- maturational changes in evaluator sensitivity
- variation in pattern fields across cultures and environments.

Modelling how hierarchical pattern modelling unfolds under such constraints should refine the theory and improve its explanatory scope.

A.8 Pattern density and stability

The main text treats pattern density as a qualitative property that contributes to stickiness. This section sketches a minimal formalisation that is compatible with the objectives in A.1 and the evaluator structure in A.4.

Let a pattern consist of identifiable subpatterns .

A.8.1 Structural, evaluative and field components

Define three components.

Structural connectivity

$$C(h) = \frac{1}{n} \sum_{i,j} w_{ij}$$

Evaluator reinforcement

$$E(h) = \sum_k e_k(h)$$

Field amplification

$$F(h) = \lambda_h$$

Define pattern density as a weighted combination of these components:

$$D(h) = \alpha C(h) + \beta E(h) + \gamma F(h)$$

A.8.2 Density, loss and stability

The loss term from A.1 measures epistemic fit and complexity. Density provides an additional property that influences whether a pattern persists despite its loss.

Define a simple stability function:

$$S(h) = \sigma \big(\eta D(h) - \delta L(h \mid x_{1:T}) \big)$$

is a logistic squashing function,

and are scaling parameters.

High density can compensate for moderate epistemic loss, which yields sticky but inaccurate patterns. Low density patterns require low loss in order to remain stable.

A.8.3 Relation to the Meta Pattern Rule

In the Meta Pattern Rule dynamics in Appendix D, each pattern has a total score that influences its weight . Pattern density can be incorporated as a slowly varying bias term:

$$\text{Total}_i(t) \leftarrow \text{Total}_i(t) + \kappa_D D(h_i)$$

Under this modification:

patterns with high epistemic fit and high density gain the strongest advantage

patterns with high density but poor fit can persist for some time, which matches the main text account of superstition and misinformation

patterns with low density and poor fit are rapidly pruned

This extension keeps the original objectives and evaluator structure intact, while making explicit how internal structure, evaluators and pattern fields jointly influence long term stability. It also suggests measurable quantities, such as estimates of connectivity, evaluator strength and field activation, that could be used in empirical tests of pattern stickiness.

Appendix B. Indicative sources and intellectual lineage

This appendix lists representative sources that inform the hierarchical pattern modelling hypothesis. It is organised by theme rather than exhaustively, to keep the main text readable while making the intellectual lineage explicit.

B.1 Predictive processing and the Bayesian brain

These works ground the idea of the brain as a hierarchical prediction machine that minimises prediction error or surprise. Clark, A. (2013). Whatever next. Predictive brains, situated agents, and the future of cognitive science. Behavioral and Brain Sciences. Friston, K. and colleagues. Bayesian brain and active inference work. introductory summaries such as Cocks (2018) give non technical overviews. Piekarski, M. (2021). Understanding Predictive Processing. A review. outlines core claims and theoretical difficulties in predictive processing. Mangalam, M. et al. (2025). The myth of the Bayesian brain. critical perspective on the scope and limits of Bayesian

framing. These sources relate most directly to Sections 2 and 8, where the paper discusses hierarchical prediction and connects to predictive processing without adopting all of its commitments.

B.2 Bayesian models of cognitive development and causal learning

This cluster underpins statements about children as hypothesis testers, structured representations of causality and the role of priors in learning. Representative sources include work on Bayesian networks and causal learning in development, for example:

Gopnik, A. and Tenenbaum, J. (2007). Bayesian networks, causal learning and development. An overview of Bayesian approaches to early causal learning.

Further work by Griffiths, Tenenbaum, Schulz, Gopnik and colleagues on Bayesian models of causal induction and concept learning supports the idea that children perform structured probabilistic inference rather than simple association.

These works inform Section 2 on innate structure, Section 7 on learning and transfer, and the general claim that human learners approximate structured probabilistic inference rather than simple association.

B.3 Curiosity, information seeking and the Goldilocks effect

These sources connect to the treatment of curiosity, boredom and evaluators in Section 7.
Kidd, C., Piantadosi, S. and Aslin, R. (2012). The Goldilocks effect. Human infants allocate attention to visual sequences that are neither too simple nor too complex. PLoS One. Kidd, C. and Hayden, B. (2015). The psychology and neuroscience of curiosity. review of curiosity as information seeking under uncertainty. van Lieshout, L. F. F. et al. (2020). Why so curious. Quantifying mechanisms of information seeking. Recent work on infants' moment to moment information sampling and intermediate information content. for example Altmann et al. (2025). These papers support the idea that evaluators favour patterns that improve prediction in intermediate difficulty regimes rather than pure extremes.

B.4 Expertise, chunking and pattern compression

These sources anchor claims about expertise as pattern compression and structured memory, as discussed in Section 7.
Chase, W. G. and Simon, H. A. (1973). Perception in chess. classic chunking account of expert memory.
Gobet, F. and Simon, H. A. (1998). Pattern recognition makes search possible. elaboration into template theory.
Gobet, F. (1997, 1998 and later). Work comparing theories of expertise and developing CHREST style models. These models illustrate how large libraries of structured chunks or templates can support rapid skilled performance, aligning with the paper's treatment of hierarchical patterns and compression.

B.5 Analogy, structure mapping and transfer

These works inform the discussion of transfer, structural alignment and far generalisation in Sections 7 and 9.
Gentner, D. (1983). Structure mapping. A theoretical framework for analogy.
Schumacher, R. M. and Gentner, D. (1988). Transfer of training as analogical mapping. Gentner and colleagues on analogical encoding and learning, for example work synthesised in later overviews of learning and transfer. These sources show that transfer often depends on alignment of deeper relational structure, not just surface features.

B.6 Heuristics, bias and illusory truth

This cluster underpins the treatment of bias, superstition, evaluator distortions and misinformation in Sections 4, 6 and 7. Representative sources include:

Tversky, A. and Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases.
Kahneman, D. (2011). Thinking, Fast and Slow.
Ecker, U. K. H. et al. (2022). The Psychological Drivers of Misinformation Belief and Its Resistance to Correction.

Work on the illusory truth effect and repetition, for example Hasher, Goldstein and Toppino, shows how repeated exposure can increase perceived truth in the absence of improved epistemic fit. In HPM terms, repetition strengthens pattern weights and evaluator signals in a local pattern field, stabilising false patterns despite poor epistemic fit.

B.7 Social learning, norms and pattern fields

These sources relate to the notion of pattern fields, social evaluators and group level pattern regimes in Sections 4, 6 and 9.
Boyd, R. and Richerson, P. (1985 onward). Cultural evolution and social learning models, including conformist transmission.
Henrich, J. and colleagues on social learning, conformity and cumulative culture. for example Muthukrishna, Morgan and Henrich (2016) on when and who to copy. Recent work on conditions for cumulative cultural evolution and conformity, for example Wakano, Aoki and others. These models support treating institutions, media environments and peer groups as structured pattern fields that shape which patterns and evaluators stabilise in populations.

B.8 Education, scaffolding and developmental support

These works ground the discussion of education as managed exposure to structure and evaluator shaping in Sections 7 and 9.
Vygotsky, L. S. foundational ideas on the Zone of Proximal Development, summarised and extended in Shabani (2010) and later instructional analyses. Bruner, J. on scaffolding and guided learning, with modern summaries by educational theorists linking scaffolding to ZPD. Contemporary overviews of scaffolding practice that emphasise staged withdrawal of support as competence grows. These sources link the abstract pattern language to practical questions about sequencing tasks, providing feedback and designing curricula.

B.9 Computational and information theoretic perspectives on learning

These references provide broader context for viewing learning as compression, model selection and structure discovery. Tenenbaum, J. B. and colleagues. Bayesian models of cognition chapter, as cited above, covering causal structure learning, category learning and word learning as probabilistic inference. General introductions to Bayesian models in cognitive science, such as Perfors et al. (2011). Reviews of predictive processing, such as Piekarski (2021), which discuss conceptual overlaps and differences with other computational frameworks. These works help situate the hierarchical pattern modelling hypothesis within existing probabilistic and information theoretic approaches rather than outside them.

Appendix C. Reviewer-Facing Clarifications

This appendix provides a concise justification of conceptual precision, grounding in established research, points of genuine novelty and the falsifiable character of the hierarchical pattern modelling hypothesis. It is intended to assist reviewers in situating the contribution relative to existing frameworks.

C.1 Conceptual precision and avoidance of vagueness

The framework avoids vagueness by placing explicit constraints on what counts as a pattern. A representation must have internal coherence, support prediction or action, and be maintained by measurable evaluators. These criteria exclude interpretations in which all mental contents qualify equally. The distinction between pattern substrates, dynamics, evaluators and fields introduces operational structure that can be mapped directly to observable measures such as prediction error, structural compression, affective load, social reward and learning trajectories. This provides a non metaphorical vocabulary for describing where patterns reside, how they change and why some stabilise while others decay.

C.2 Relationship to established research

Several components of the hypothesis map directly to well established literatures. Predictive processing and Bayesian learning provide hierarchical prediction models that align with the framework’s treatment of latent structure and prediction error. Bayesian models of cognitive development already interpret children as hypothesis testers discovering structured generative rules. Research on curiosity shows preferential engagement with intermediate complexity regimes, consistent with the evaluator account proposed here. Expertise research demonstrates pattern compression and transfer across surface variation. Work on heuristics, illusory truth and social learning supports the claim that affective and social evaluators can stabilise false patterns. These mappings show that the hypothesis integrates, rather than replaces, current models.

C.3 Novel contributions

The novelty lies in unifying diverse mechanisms within a single architecture and introducing distinctions that do not appear together in existing theories.

1. The four role structure of substrates, dynamics, evaluators and fields provides a clear separation between where patterns are encoded, how they are updated, what selects them and how environments shape their availability.
2. Evaluators are elevated to central organising components. This provides a principled account of why incorrect patterns can be stable and why identity linked beliefs and affective distortions persist.
3. Pattern fields generalise learning environments to include media systems, institutions and scientific practices. This extends cognitive accounts to collective and cultural levels without altering the underlying machinery. These additions allow the hypothesis to connect micro level learning, group divergence and institutional design within a single vocabulary.

C.4 Falsifiability and predictive constraints

The theory makes directional behavioural predictions that distinguish it from simpler associative or purely predictive error models. These include: expected asymmetries between deep structural changes and surface changes; compression and cross-surface transfer as signatures of expertise; selective stabilisation of false patterns under affective load; preferential exploration of intermediate structure in curiosity; group convergence depending on field quality; and far-transfer advantages from pattern-centred instruction.

Each prediction specifies a counter pattern. If learners prove equally sensitive to surface and structural changes, if expertise does not exhibit compression, if affective load does not increase resistance to correction, or if field design does not alter convergence rates, the stronger claims of the hypothesis would be weakened. This provides a clear falsifiable backbone rather than an unfalsifiable narrative.

C.5 Summary

This appendix clarifies how the hierarchical pattern modelling framework achieves conceptual precision, integrates with established literatures, offers identifiable points of novelty and yields falsifiable predictions. It is intended to assist reviewers in assessing the coherence and empirical tractability of the hypothesis.

Appendix D. Minimal Mathematical Formulation of the Meta Pattern Rule

Working Draft. Minimal, Derived, and Subject to Revision

This appendix provides a minimal formal structure that captures the core commitments of the hierarchical pattern modelling hypothesis. The aim is not to specify a complete implementation but to show that evaluator weighted pattern selection is mathematically tractable.

D1. Pattern Representation

Each pattern hih_ihi is represented as a simple generative model:

pi(x1:T∣θi)p_i(x_{1:T} \mid \theta_i)pi(x1:T∣θi)

with parameter vector θi\theta_iθi.
Full HPM would incorporate multilevel generative structure. Here the model remains minimal.

D2. Epistemic Loss and Accuracy

For a data stream x1:Tx_{1:T}x1:T, define instantaneous loss:

ℓi(t)=−logpi(xt∣hi).\ell_i(t) = -\log p_i(x_t \mid h_i).ℓi(t)=−logpi(xt∣hi).

Define a running loss as an exponential moving average:

Li(t)=(1−λL)Li(t−1)+λL ℓi(t),L_i(t) = (1 - \lambda_L) L_i(t-1) + \lambda_L \, \ell_i(t),Li(t)=(1−λL)Li(t−1)+λLℓi(t),

with 0<λL≤10 < \lambda_L \leq 10<λL≤1 and some initial Li(0)L_i(0)Li(0).

Define accuracy as:

Ai(t):=−Li(t).A_i(t) := -L_i(t).Ai(t):=−Li(t).

Accuracy increases when the running loss decreases.

D3. Evaluators

HPM assumes multiple evaluators. In this minimal form:

- Epistemic evaluator (accuracy)
  Eiepi(t):=Ai(t).E^{\text{epi}}_i(t) := A_i(t).Eiepi(t):=Ai(t).
- Affective evaluator
  Eiaff(t).E^{\text{aff}}_i(t).Eiaff(t).
- Social evaluator
  Eisoc(t).E^{\text{soc}}_i(t).Eisoc(t).

To prevent double counting of accuracy, the combined evaluator excludes the epistemic term:

Ji(t):=β Eiaff(t)+γ Eisoc(t),J_i(t) := \beta \, E^{\text{aff}}_i(t) + \gamma \, E^{\text{soc}}_i(t),Ji(t):=βEiaff(t)+γEisoc(t),

with non-negative weights β,γ\beta, \gammaβ,γ.

D4. Total Pattern Score

Define the total score that determines selective advantage:

Totali(t):=Ai(t)+Ji(t).\text{Total}_i(t) := A_i(t) + J_i(t).Totali(t):=Ai(t)+Ji(t).

Patterns with higher accuracy and higher evaluator reward gain weight.

D5. Meta Pattern Rule Dynamics

Let wi(t)w_i(t)wi(t) be the weight of pattern hih_ihi at time ttt. Weights evolve through a discrete time replicator style update with conflict inhibition:

wi(t+1)=wi(t)w_i(t + 1) = w_i(t)wi(t+1)=wi(t)
+η(Totali(t)−Total¯(t)) wi(t)\quad + \eta \big(\text{Total}_i(t) - \bar{\text{Total}}(t)\big) \, w_i(t)+η(Totali(t)−Total¯(t))wi(t)
−βc∑j≠iκij wi(t)wj(t),\quad - \beta_c \sum_{j \neq i} \kappa_{ij} \, w_i(t) w_j(t),−βc∑j=iκijwi(t)wj(t),

where

Total¯(t):=∑jwj(t) Totalj(t)\bar{\text{Total}}(t) := \sum_j w_j(t) \, \text{Total}_j(t)Total¯(t):=∑jwj(t)Totalj(t)

is the population average total score, η\etaη is a learning rate, βc\beta_cβc is a conflict scale and κij\kappa_{ij}κij is an incompatibility matrix. This produces evaluator weighted selection, pattern coexistence and inhibition of incompatible patterns.

D6. Minimal Pattern Field Influence

Pattern fields model social and environmental structure.
A simple frequency based field influence is:

Fi(t)=ρ⋅freqi(t),F_i(t) = \rho \cdot \text{freq}_i(t),Fi(t)=ρ⋅freqi(t),

and

Totali(t)←Totali(t)+γfFi(t).\text{Total}_i(t) \leftarrow \text{Total}_i(t) + \gamma_f F_i(t).Totali(t)←Totali(t)+γfFi(t).

A full model would incorporate structural and temporal dynamics.

D7. Hierarchical Form (Minimal Extension)

For hierarchical patterns with latent variables zt(2),zt(1)z_t^{(2)}, z_t^{(1)}zt(2),zt(1), let qi(t)q_i(t)qi(t) be an approximate posterior over latent trajectories for pattern hih_ihi at time ttt. Define the hierarchical loss as a negative evidence lower bound:

Lihier(t):=Eqi(t)[logqi(t)(z(1),z(2))]−Eqi(t)[logpi(x,z(1),z(2))].L_i^{\text{hier}}(t) := \mathbb{E}_{q_i(t)}[\log q_i(t)(z^{(1)}, z^{(2)})] - \mathbb{E}_{q_i(t)}[\log p_i(x, z^{(1)}, z^{(2)})].Lihier(t):=Eqi(t)[logqi(t)(z(1),z(2))]−Eqi(t)[logpi(x,z(1),z(2))].

Define compression at time ttt as the mutual information between latent layers under qi(t)q_i(t)qi(t):

Compi(t):=Hqi(t)[z(1)]−Hqi(t)[z(1)∣z(2)].\text{Comp}_i(t) := H_{q_i(t)}[z^{(1)}] - H_{q_i(t)}[z^{(1)} \mid z^{(2)}].Compi(t):=Hqi(t)[z(1)]−Hqi(t)[z(1)∣z(2)].

Define the hierarchical total score as:

Totalihier(t):=−Lihier(t)+βcomp Compi(t)+Ji(t),\text{Total}_i^{\text{hier}}(t) := - L_i^{\text{hier}}(t) + \beta_{\text{comp}} \, \text{Comp}_i(t) + J_i(t),Totalihier(t):=−Lihier(t)+βcompCompi(t)+Ji(t),

where βcomp≥0\beta_{\text{comp}} \ge 0βcomp≥0 weights the contribution of compression and Ji(t)J_i(t)Ji(t) is the non epistemic evaluator term from D3. When z(2)z^{(2)}z(2) collapses, this reduces to the flat case.

D8. Limitations

This minimal model omits:

- structural change
- abstract–concrete transitions
- forgetting
- evaluator learning
- dynamic pattern fields

It provides a tractable base case for testing core HPM predictions.

Appendix E. Minimal Formalisation of Structural Recombination

Working Draft. Minimal, Derived, and Consistent with Appendix D

Structural recombination introduces a mechanism for generating new patterns from existing ones before they enter the Meta Pattern Rule dynamics.

E1. Pattern Structure

Each pattern hhh is a three level generative model:

h={p(2),p(1),p(0),θ},h = \{p^{(2)}, p^{(1)}, p^{(0)}, \theta\},h={p(2),p(1),p(0),θ},

with latent hierarchy:

zt(2)→zt(1)→xt.z_t^{(2)} \rightarrow z_t^{(1)} \rightarrow x_t.zt(2)→zt(1)→xt.

Factorisation:

ph(x1:T,z1:T(1),z1:T(2))=p_h(x_{1:T}, z_{1:T}^{(1)}, z_{1:T}^{(2)}) =ph(x1:T,z1:T(1),z1:T(2))= p(2)(z1(2)) p(1)(z1(1)∣z1(2)) p(0)(x1∣z1(1))\quad p^{(2)}(z_1^{(2)}) \, p^{(1)}(z_1^{(1)} \mid z_1^{(2)}) \, p^{(0)}(x_1 \mid z_1^{(1)})p(2)(z1(2))p(1)(z1(1)∣z1(2))p(0)(x1∣z1(1))

×∏t=2Tp(2)(zt(2)∣zt−1(2)) p(1)(zt(1)∣zt(2)) p(0)(xt∣zt(1)).\quad \times \prod_{t=2}^T p^{(2)}(z_t^{(2)} \mid z_{t-1}^{(2)}) \, p^{(1)}(z_t^{(1)} \mid z_t^{(2)}) \, p^{(0)}(x_t \mid z_t^{(1)}).×∏t=2Tp(2)(zt(2)∣zt−1(2))p(1)(zt(1)∣zt(2))p(0)(xt∣zt(1)).

This formulation explicitly separates the initial state (t=1)(t = 1)(t=1) from the subsequent transitions (t=2 to T)(t = 2 \text{ to } T)(t=2 to T), avoiding any reference to an undefined initial latent state z0(2)z_0^{(2)}z0(2).

E2. Recombination Operator

Define a recombination operator:

R:H×H→H,\mathcal{R} : \mathcal{H} \times \mathcal{H} \rightarrow \mathcal{H},R:H×H→H,

subject to structural constraints C\mathcal{C}C.

Given parent patterns ha,hb∈Hh_a, h_b \in \mathcal{H}ha,hb∈H, recombination produces:

h∗=R(ha,hb∣C).h^* = \mathcal{R}(h_a, h_b \mid \mathcal{C}).h∗=R(ha,hb∣C).

E3. Minimal Recombination Rules

In the minimal model the operator may combine:

- structural elements of p(2),p(1),p(0)p^{(2)}, p^{(1)}, p^{(0)}p(2),p(1),p(0)
- latent transitions
- evaluator relevant priors
- compression relevant features

This defines a space of candidate recombinations without fixing a single implementation.

E4. Insight Evaluator

Insight quantifies novelty and effectiveness of a recombined pattern:

I(h∗)=βorig(αnov Nov(h∗)+αeff Eff(h∗)),I(h^*) = \beta_{\text{orig}} \left(\alpha_{\text{nov}} \, \text{Nov}(h^*) + \alpha_{\text{eff}} \, \text{Eff}(h^*) \right),I(h∗)=βorig(αnovNov(h∗)+αeffEff(h∗)),

where Nov(h∗)\text{Nov}(h^*)Nov(h∗) measures structural novelty and Eff(h∗)\text{Eff}(h^*)Eff(h∗) measures performance on relevant tasks.

Insight boosts the affective evaluator for the new pattern:

Eh∗aff(t0)←Eh∗aff(t0)+I(h∗),E^{\text{aff}}_{h^*}(t_0) \leftarrow E^{\text{aff}}_{h^*}(t_0) + I(h^*),Eh∗aff(t0)←Eh∗aff(t0)+I(h∗),

at the time t0t_0t0 when h∗h^*h∗ is first created.

E5. Entry into Meta Pattern Rule Dynamics

The new pattern enters the population with initial weight:

wh∗(t0)=κ0I(h∗).w_{h^*}(t_0) = \kappa_0 I(h^*).wh∗(t0)=κ0I(h∗).

It then evolves according to the Meta Pattern Rule dynamics in Appendix D.
If h∗h^*h∗ maintains low hierarchical loss and high compression, the temporary insight reward yields a strong initial score, enabling it to compete with and possibly displace its parent patterns ha,hbh_a, h_bha,hb.

E6. Constraint Satisfaction

The recombined pattern is retained only if it satisfies structural constraints:

C(h∗)=1.\mathcal{C}(h^*) = 1.C(h∗)=1.

If the constraint is violated, the pattern is effectively discarded:

wh∗(t)≡0.w_{h^*}(t) \equiv 0.wh∗(t)≡0.

This prevents structurally incoherent or unstable recombinations from entering the selection dynamics.